

Raju J S

System Engineer at TCS (Python, Data Engineer and ETL Developer)

BENGALURU, KARNATAKA | +91 7676443583 | RAJUJS.RAJ@GMAIL.COM | **LinkedIn:** <https://www.linkedin.com/in/raju-j-s-0849a1260>

PROFESSIONAL SUMMARY

Experienced **5.5+ years** of Software Developer with a strong background in **Python Backend, Flask-based API development, Database management, ETL Development, Data Analysis and Production Support**. Seeking a challenging role in the IT industry where I can design, develop, and optimize secure, scalable applications, contribute to system architecture and performance improvements, and continuously enhance my technical expertise while delivering high-quality software solutions.

TECHNICAL SKILLS

Programming Languages : Python, SQL

API's Frameworks : Flask, FastApi.

Database : MSSQL, PostgreSQL, MongoDB, MySql

Tools & Technologies: Docker, Git, GitHub, Nginx, Postman, Jira, Artifactory

Data & ETL : Pandas, Data Analysis, ETL Development, Tableau

Concepts: REST APIs, Microservices, API Security (JWT), Database Design, Debugging, Unit Testing (pytest), Swagger UI, Production Support and fixes the issue.

PROFESSIONAL EXPERIENCE

Python Backend and API's Development :

- Developed **scalable Backend** and **REST APIs** using Flask and FastAPI, improving system response time by ~30%
- Designed scalable backend architecture for high-performance applications.
- Built backend systems using **microservices architecture**.
- Developed dashboards and analytics using **Pandas DataFrames**.
- Implemented **JWT** based authentication and authorization.
- Built secure API endpoints with custom decorators and middleware.
- Custom headers and handled secure API communication.
- Implemented input validation and error handling mechanisms.
- Designed alert systems with **Email and SMS notifications**.

Database & SQL :

- Designed and optimized SQL queries for MSSQL / MySQL.
- Designed and implemented **ETL pipelines** using MSSQL, ensuring reliable data transformation
- Optimized complex SQL queries, improving database performance and reducing execution time
- Handled NULL conditions, filtering, and pattern matching using LIKE and regex.
- Designed normalized database schemas.
- Create a database design, including CRUD operations and Implemented complex queries using joins, filtering, sorting, and indexing.

Testing, Debugging, Deployment & Production Support :

- Wrote unit test cases using pytest and mock and Improved code quality using coverage reports.
- Reduced production issues via proactive monitoring and debugging
- Containerized applications using **Docker**, improving deployment efficiency
- Performed API testing using Postman.

- Used logging and debugging tools for issue resolution.
- I worked on Git for version control.
- Managed environment configurations.
- Configured **Nginx as reverse proxy** for secure routing
- Resolved high-priority production issues through **root cause analysis**, improving system stability

Data Analysis & Visualization :

- Created interactive dashboards and visualizations using libraries like Pandas, Matplotlib and Tableau.

PROJECTS

Tata consultancy services Pvt Ltd | System Engineer

12 Oct 2022 to present

Project :- Client Compass Application | Fee Transformation | Analytics and Risk Monitor (ARM1.0 & AMR2.0)

- Developed a centralized platform for **Relationship Managers (RM)** to monitor client financial data in a single interface
- Built backend services to **aggregate and process client data** for risk analysis, overdue balances, and loan account tracking
- Implemented **risk assessment logic** to identify potential financial risks and highlight critical accounts
- Enabled automated **alerts and reminders** for RMs to review high-risk clients and pending actions
- Designed and optimized **ETL workflows** to ensure accurate and timely data processing
- Improved decision-making by providing **real-time insights and consolidated client views**

IDP Education india service LLP | Associate Application Developer

04 Aug 2022 to 21 Sep 2022

Project :- Student Portal Application for IELTS exam.

- Integrated **Facebook API** for user authentication and platform features
- Contributed to backend enhancements for student portal applications

Fujitsu Consulting India Pvt Ltd | Developer Analyst

24 Mar 2022 to 02 Aug 2022

- Worked on **Python-based application development and training**
- Strengthened backend development and debugging skills

Elmeasure India Private Limited | Associate Software Engineer

10 Feb 2020 to 10 Mar 2022

Project :- Energy management v7 and Energy management v8

- Developed a real-time **energy monitoring and analytics platform** to track and control power consumption using smart meters
- Processed raw meter data including **voltage, current, power, watts, amps, and phase parameters** for detailed analysis
- Integrated **MQTT protocol** to subscribe and ingest real-time data streams via MQTT broker
- Built backend data processing using **Python and Pandas** to transform raw data into actionable insights
- Designed and implemented **interactive dashboards and reports** (hourly/daily/monthly) based on configurable parameters
- Implemented **alerting system** to trigger email/SMS notifications when energy parameters exceeded defined thresholds, improving safety and monitoring
- Optimized data pipelines for **efficient real-time processing and reporting**

EDUCATION

Bachelors of Engineer in Computer Science and Engineer

July 2015 - July 2019

S T J Institute of Technology, Ranebennur, Karnataka (VTU University)